



The Recursive Theory of Knowledge Augmentation: Integrating human intuition and knowledge in Artificial Intelligence to augment organizational knowledge

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Abstract

Artificial intelligence (AI) has increased the ability of organizations to accumulate tacit and explicit knowledge to inform management decision-making. Despite the hype and popularity of AI, there is a noticeable scarcity of research focusing on AI's potential role in enriching and augmenting organizational knowledge. This paper develops a recursive theory of knowledge augmentation in organizations (the KAM model) based on a synthesis of extant literature and a four-year revised canonical action research project. The project aimed to design and implement a human-centric AI (called Project) to solve the lack of integration of tacit and explicit knowledge in a scientific research center (SRC). To explore the patterns of knowledge augmentation in organizations, this study extends Nonaka's SECI (socialization, externalization, combination, and internalization) model by incorporating the human-in-the-loop Informed Artificial Intelligence (IAI) approach. The proposed design offers the possibility to integrate experts' intuition and domain knowledge in AI in an explainable way. The findings show that organizational knowledge can be augmented through a recursive process enabled by the design and implementation of human-in-the-loop IAI. The study has important implications for research and practice.

Keywords Informed Artificial Intelligence · Human-Centric AI · Human-in-the-loop AI · Explainable AI · Dynamic Theory of Organizational Knowledge Creation · Canonical action research · Knowledge augmentation

1 Introduction

AI for organizational use has been studied across various academic disciplines since the 1950s (Cardenas, 1977; Dewhurst & Gwinnett, 1990; Lawrence, 1991). AI has significantly impacted decision-making in organizations (Dennehy et al., 2022; Duan et al., 2019; Shrestha et al., 2021; von Krogh, 2018). According to Fosso Wamba and Queiroz (2021), the most popular word in AI research is "Knowledge." Indeed, AI accumulates knowledge that directly guides managers' decisions (Collins et al., 2021;

Faraj et al., 2018; Furman & Teodoridis, 2019) and helps gain new insights to augment human knowledge (Cockburn et al., 2019; Sahay et al., 2021; Shollo & Galliers, 2016).

Despite the long history of interest in AI, academic research has focused mainly on the technical aspects of the technology (Rokach, 2010), decision-support (Shrestha et al., 2021), explainable AI (Arrieta et al., 2020; Johnson et al., 2021a), ethical AI (Dennehy et al., 2021; Enholm et al., 2021; Tutun et al., 2022), and responsible AI (Johnson et al., 2021b, 2022; Liu et al., 2021; Merhi, 2022). Whereas relatively less research examines how new organizational knowledge may be obtained through AI (Cockburn et al., 2019; Furman & Teodoridis, 2019). This is problematic as there is a limited understanding of how the design and implementation of an AI can aggregate tacit and explicit knowledge to augment organizational knowledge.

This study addresses this issue by developing a new theory to explain the recursive aspects of knowledge augmentation in organizations. The Recursive Theory of Knowledge Augmentation (the KAM model) explains how tacit and explicit knowledge held by individuals and organizations

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can be simultaneously enlarged and enriched through the recursive process enabled by human-in-loop Informed AI. The human-in-the-loop Informed AI is a new form of AI that emerged recently. This new form of AI is a Human-Centric AI that incorporates Machine Learning bottom-up methods with knowledge-guided top-down methods (Grønsund & Aanestad, 2020; Johnson et al., 2022; Pan, 2016; Von Rueden et al., 2019; Zhuang et al., 2017).

The research reports on the results of a four-year canonical action research (CAR) project (2017–2020) conducted at a leading French archaeological research center (SRC). The SRC researches human history, evolution, and culture and aims to explain temporal changes in human societies by reconstructing past living conditions. The CAR project was designed to introduce an Informed AI at the SRC to recreate the Palaeolithic living conditions of the Man cave.¹ The project aims to explain prehistoric human behavior by augmenting SRC's organizational knowledge.

As suggested by Davison et al. (2012), two theoretical streams are drawn: 1) A focal theory to provide the intellectual basis for the action-oriented change, and 2) an instrumental theory to play a critical role in helping to ensure that the focal theory is relevant to practice, and that practice is connected to the focal theory (Davison et al., 2012). Therefore, to explore the patterns of knowledge augmentation in organizations, our research draws on the seminal work of Nonaka (1994) on organizational knowledge. Introduced as focal theory, the Dynamic Theory of Organizational Knowledge Creation can disambiguate the link between AI use and organizational knowledge creation. The second stream tracks the recent approach of Human-Centric AI (Ramchurn et al., 2021) and specifically Informed Artificial Intelligence (Johnson et al., 2022; Karpatne et al., 2017; Pan, 2016; Von Rueden et al., 2019; Zhuang et al., 2017). This approach is based on integrating human knowledge in the data-driven machine learning process. Based on the above theoretical underpinning, this paper reflects on developing a Recursive Theory for Knowledge Augmentation. The proposed theory identifies a straightforward recursive process that facilitates the dynamic creation of valuable organizational knowledge augmented through IAI implementation. Five new modes of knowledge augmentation were identified: 1) Alimentation, 2) Aggregation, 3) Reflection, 4) Amplification, and 5) Fusion.

The remainder of the paper is structured as follows: Nonaka's Dynamic Theory of Knowledge Creation is discussed in the context of knowledge augmentation, and human-in-the-loop Informed Artificial Intelligence. Then,

the research method, data collection, and analysis techniques are discussed. Next, the research results are presented, followed by a discussion, limitations, and directions for future research. The paper ends with a conclusion.

2 Literature Review

In organization theories, a knowledge-based perspective was developed, extending the resource-based theory of the firm. The knowledge-based perspective postulates that the firm can effectively apply its knowledge to create new knowledge and act based on it to achieve a competitive advantage (Alavi & Leidner, 2001). Given the importance of the process of knowledge creation, the objective of the theoretical part is to synthesize the relevant theories related to organizational knowledge and combine them with the Human-Centric AI perspective. The idea of knowledge conversion and creation may be traced from Anderson's Adaptive Control of Thought Model (Anderson, 1983), developed in cognitive psychology, and its improvement presented by Nonaka (1994) in the dynamic theory of organizational knowledge creation. The Human-Centric perspective and its human-in-the-loop approach aim to improve AI performance, robustness, fairness, accountability, transparency, and explainability (Horvatić & Lipić, 2021). Integrating these two theories can enrich our understanding of the organizational process of knowledge creation in the AI era.

2.1 The Focal Theory: The Dynamic Theory of Organizational Knowledge Creation

The concept of knowledge is an old complex concept (Davenport & Prusak, 1998). Since Plato, defining knowledge has occupied philosophers' minds and led to many epistemological debates (Alavi & Leidner, 2001). The paper adopts the definition of Davenport & Prusak (1998), which considers knowledge as a fluid mix of framed experiences, values, context information, and expert insight that provides a framework for evaluating and incorporating new experiences and information. To distinguish knowledge from data and information, this paper adopts the commonly held view of data as raw numbers and facts. Information is created when data are processed, organized, or structured to provide context and meaning. Knowledge is created when information is converted to solve problems (Alavi & Leidner, 2001). Knowledge exists at the individual and the collective or social levels (Nonaka, 1994). *Individual knowledge* is created by and inherent in the individual. *Social knowledge* is created by and inherent in the collective action of a group.

Knowledge is dynamically created in social interaction amongst individuals through conversion between explicit and tacit knowledge (Nonaka, 1994). *Explicit knowledge* can

¹ The Project made possible the study of the behavior of prehistoric man who lived in a cave in southern France between 600,000 and 90,000 years ago.

be expressed in formal, systematic language and shared, for instance, in data, scientific formulas, or specification manuals (Nonaka et al., 2000). In contrast, influenced by philosopher Polanyi (1966), Nonaka considers tacit knowledge deeply rooted in action, commitment, and involvement in a specific context.

To illustrate the dynamic of knowledge creation and conversion, Nonaka proposed the famous SECI (socialization, externalization, combination, and internalization) process of knowledge creation and conversion. In this perspective, knowledge creation forms an infinite spiral based on an interaction between individuals. Indeed, tacit knowledge is created at the individual level. Organizations amplify their conversion spiral through four modes: socialization, externalization, combination, and internalization. The conversion from tacit knowledge to tacit knowledge is presented as *socialization*. It is the process of creating tacit knowledge through shared experience. It is developed through observation, imitation, and practice. As for converting tacit knowledge into explicit knowledge, Nonaka (1994) coins this as the process of *externalization*. Metaphors and analogies can enable team members to articulate their perspectives and reveal hidden tacit knowledge that is otherwise hard to communicate.

On the one hand, metaphors help individuals infer from a model of different behavior by learning through symbols and describing the world in terms of prototypes. Conversely, analogy reduces ambiguity by allowing the functional operation of new concepts referring to things already understood. The *combination* is the process of creating explicit knowledge from explicit knowledge. Individuals exchange and combine knowledge through such exchange mechanisms as meetings and telephone conversations. *Internalization* is the process of converting explicit knowledge into tacit knowledge; an iterative process of trial and error can trigger internalization through a process of learning by doing. The spiral in the SECI model indicates that each time the knowledge goes around the loop, it is enlarged and improved.

2.2 The Instrumental Theory: A Human-Centric Perspective

Nowadays, AI's increased capabilities enable organizations to address significant problems in our society (Lepri et al., 2021). Indeed, algorithms are increasingly used to support human decisions (Duan et al., 2019; Faraj et al., 2018). Recent AI success comes from implementing deep neural network algorithms in image recognition (Gu et al., 2018). Deep neural network models are data-driven AI that enables automatic feature engineering, outperforming traditional handcrafted methods in image, text, and speech recognition (Horvatić & Lipić, 2021). Nevertheless, by significantly reducing and sometimes eliminating the reliance

on domain-expert, these new data-driven approaches have raised concerns about their models' transparency (Haibe-Kains et al., 2020) and interpretability (Lipton, 2018). To fully trust AI solutions, researchers are prone to a Human-Centric explainable AI that can provide humans with understandable interpretations of their predictions (Ramchurn et al., 2021). *Human-Centric AI* can be designed to continuously collaborate and learn from human input while providing explainable and interpretative predictions. According to Horvatić and Lipić (2021), a Human-Centric AI can enable humans to control and continuously improve AI applications' performance, robustness, fairness, accountability, transparency, and explainability.

User participation in the design of Information Systems was recommended long ago. In the classical approach, this participation was reduced to analyzing the needs and then to the tests. In the Human-Centric AI approach, participation is requested at the experts' level, which is all along the process. As they know the domain well, they must intervene permanently and in all the design and implementation phases to feed the design with their experience and knowledge. So that makes this approach like the agile methods but engages the experts of the domain and not the simple users. Companies' performance can be improved when AI and humans collaborate (Shrestha et al., 2019; Wilson & Daugherty, 2018). According to Wilson and Daugherty (2018), the human role will start with the design and training of the AI. Then, it continues during the explanation and implementation. It continues in a loop to sustain the trustworthy implementation of AI. Experts can enhance AI designs by incorporating common-sense reasoning and common-sense knowledge into the design and training process (Davis & Marcus, 2015). Human-Centric AI can augment human decision-making capabilities by analyzing the possible consequences of each action and explaining the reasons behind such possible consequences.

Following this motivation, the recently emerging trend in AI design is based on exploring human-in-the-loop (Grønsund & Aanestad, 2020) approaches and developing contextual explanatory AI applications. *Human-in-the-loop* approaches involve domain experts in all steps of the AI design, from data collection and cleaning to algorithm selection, training, testing, and AI implementation and interpretation. However, the design of Human-in-the-loop AI for management decision-making is not without challenges. Indeed, designing such AI applications is a multidisciplinary effort involving multiple collaborations between data scientists, information systems specialists, and domain experts. To bring AI to its most useful state, multidisciplinary scientists must unite to establish common standards and rules for AI. Thus, integrating knowledge from the field of AI design with knowledge from the field of usage is imperative to reach the Human-Centric AI aims. This is also known in computer science as Informed AI

(Johnson et al., 2022; Von Rueden et al., 2019), theory-guided AI (Faghmous et al., 2014; Karpatne et al., 2017), or AI 2.0 (Pan, 2016; Zhuang et al., 2017).

2.3 The Advent of Human-in-the-loop Informed Artificial Intelligence

Combining the Dynamic Theory of Organizational Knowledge Creation (focal theory) with the Human-Centric AI approach can expand their corresponding assumption to the context of knowledge augmentation in organizations. Designing and implementing a human-in-the-loop Informed Artificial Intelligence (IAI) can be a strong operationalization of the Human-Centric AI approach adapted to knowledge augmentation challenges. It assumes integrating explicit and tacit knowledge into the design, training, and implementation of AI.

The human-in-the-loop Informed Artificial Intelligence (IAI) could improve AI models by explicitly incorporating domain explicit knowledge into the learning process (Von Rueden et al., 2019). Explicit knowledge comprises both object-based and rule-based knowledge (Popadiuk & Choo, 2006). Knowledge is *object-based* when codified in words, numbers, formulas or made tangible as equipment, documents, or models. Knowledge is *rule-based* when knowledge is encoded as rules, routines, and standard operating procedures. IAI is also a must when there is a need for explainable and transparent models (Johnson et al., 2021a).

Therefore, the first research proposition can be formulated as follows:

Proposition 1: Experts' object-based and rule-based knowledge can be integrated into AI design and implementation to improve the performance of the IAI models and increase their explainability.

According to Von Rueden et al. (2019), "in conventional data-driven AI, data is the input to a learning process, which detects patterns in the data. While data-driven models need a large volume of data (big data), IAI integrates human knowledge with data-driven AI to address the difficulties related to a small volume, incomplete, or missing data (Harfouche et al., 2017; Von Rueden et al., 2019).

The human-in-the-loop Informed Artificial Intelligence (IAI) could improve AI models by incorporating experts' tacit knowledge into the learning process to overcome the difficulties related to a small volume of data, incomplete data, or missing data. *Tacit knowledge* is comprised of both cognitive and technical elements. The *cognitive component* refers to an individual's mental models of mental maps, beliefs, paradigms, and viewpoints. The *technical*

component consists of concrete know-how, crafts, and skills that apply to a specific context (Alavi & Leidner, 2001).

Hence, the second research proposition can be formulated as follows:

Proposition 2: The cognitive and technical components of experts' tacit knowledge can be integrated into AI design to address difficulties related to small-volume, incomplete, or missing data.

Positioning AI solely as a set of tools automating the decision process thus limits our understanding of the impact AI might have on organizational knowledge creation. Research must consider the social and collaborative nature of knowledge creation (Shollo & Galliers, 2016). AI can produce new-to-the-world knowledge Furman and Teodoridis (2019). It might play an essential role in transforming personal knowledge into organizational knowledge and the latter into actions and learning (Shollo & Galliers, 2016).

The third research proposition can be formulated as follows:

Proposition 3: While designing and implementing an Informed AI, an organization might augment its knowledge.

This study adopts the following research methodology to answer these propositions.

3 Research Methodology

3.1 Background to the Project

To understand how human-in-the-loop IAI augments organizational knowledge, this paper analyses a four-year revised canonical action research called "Project." This project started in January 2017 and ended in November 2020. It was designed to implement a human-in-the-loop IAI at the SRC: a leading French archaeological research center. The SRC is a multidisciplinary center that studies several archaeological sites in Eurasia. Within this context, The Project was designed to bring together scientists from various domains to work on the design and the implementation of an IT artifact that links two interacting modules: 1) an ensemble-learning IAI design that integrates experts' knowledge into the data-driven machine learning process, and 2) an immersive virtual environment.

The project selected the 450,000-year-old Man Cave, a Pre-Neanderthal cave dweller, as a data source. Since the late 1960s, scientists have used statistical analysis to abstract the critical features and concepts from data collected through surveying and excavation. However, more emphasis

has been placed on gathering data & information than on the effective and efficient use of available knowledge to explain and predict the Man Cave's past behavior. Indeed, it was difficult to aggregate the large volume of tacit and explicit knowledge needed to explain Man Cave's past behavior. The data collected was not of a significant volume. Undoubtedly, analyzing archaeological data in a multidisciplinary context involves complex, non-linear relationships in which causes and effects cannot be automatically distinguished.

Consequently, addressing these challenges requires sophisticated modeling and a Human Centric design. Therefore, The Project involved many academic and professional partners in developing innovative human-in-the-loop Informed Artificial Intelligence solutions that help SRC researchers to recreate Paleolithic living conditions. The Project simulator aims to train and test (validate or reject) prehistoric human behavior hypotheses proposed by SRC researchers in a reconstructed immersive environment.

3.2 Canonical Action Research (CAR)

CAR was selected because of two reasons. First, CAR requires that researchers go through carefully designed and executed cycles of activities (Davison et al., 2004). Indeed, CAR allows researchers to understand the organizational problem in-depth to propose an adequate solution through several cycles of activities. Second, according to Davison et al. (2004), CAR requires that the process of diagnosing the problem must be continuous and involve the field actors throughout cycles. The goal is to carry out constantly evolving activities relevant to the issue that also evolves. Researchers need time to build a strong relationship with the other actors involved in the research, then observe, plan, and execute actions, considering organizational constraints and idiosyncrasies. According to Davison et al., (2004), CAR has the dual benefit of improving practice while contributing theory and knowledge within and beyond the immediate boundaries of the project. By this same logic, and even though other research models could have been used to analyze our research object in its natural context, the "design" of Susman and Evered (1978) was the most appropriate because of its interventionist and iterative approach dedicated to the development of valuable knowledge for research and practice. In addition, the research model of Susman and Evered (1978) is a canonical form of action research and one of the most widely used action research methods in the social sciences (Davison et al., 2004). Five research phases are iterated (see Table 1): (1) diagnosis, which consists in identifying the problem to be solved; (2) action planning of alternative solutions to solve the problem; (3) taking actions that correspond to the selection of solutions; (4) evaluation of the consequences of remedial actions; and (5) the

accuracy of the learning and the results of the general findings that result from each cycle.

3.3 Research Methodology: A 4-year CAR

Two authors of this paper were directly implicated in The Project: One was a team leader and participated in the project's submission. He also attended all the meetings. The other author was involved in the project from the start till the end but only in the AI part. Throughout the 4-year action research, authors drew on several data sources: internal documents, informal and conversational interviews with eight team members (22 in total, lasting between 5 and 120 min), and participant observations during which we noted verbal and non-verbal communications. Therefore, the analysis includes the meeting reports of the steering committees, the official project reports (one mid-term and one at the end), all the emails that were exchanged, all the SRC meetings (meetings were recorded and then analyzed), the consortium seminars (each lasted two days / 17 in total). For the immersion part, an analysis of the interactions of archaeologists with the virtual tool was carried out around 16 videos and eight individual interviews. The latter were transcribed and then coded using Nvivo. Researchers also focused on artifacts, facial expressions, body language analysis, pauses, sighs, stammering, and laughs when transcribing the primary statements.

On the other hand, when the data collection was completed, each respondent was assigned pseudonyms during the partial transcription of the interviews and discussions. Next, specific codes and statements expressed were categorized, matching them to the data collected during the interviews and informal discussions that support each of the classifications. Finally, the final analysis was written, starting with the classifications that permeate most participant observations and discussions.

4 Analysis and Findings

4.1 The Project's Four Cycles

When this investigation started, IAI designers did not have a precise idea of the problems and the research challenges associated with these problems. These challenges appeared over the cycles carried out over four years (see Table 1).

4.1.1 Cycle 1: The Dream of an AI

At the beginning of the project (January 2017 till March 2017), the aim was to integrate different datasets (from different sources and domains) in an AI to make firm predictions. For example, during the first seminar, an SRC

Table 1 Action research design

	First cycle (01/17 – 03/17) "The dream of an AI"	Second cycle (04/17 – 12/17) "The AI deception"	Third cycle (01/18 – 04/19) "An IAI to respond to simple questions."	Fourth cycle (05/19 – 11/20) "An integrated AI for researchers"
Diagnosing	<i>Objectives:</i> Integrating different datasets to make firm predictions <i>Problems:</i> 1) Data was not cleaned and tagged 2) Datasets had missing values and few observations for many attributes 3) Little volume of data <i>Means:</i> Designing an AI that can predict the prehistoric person's behavior <i>Sources:</i> SRC researchers: IS researchers helped archeological researchers to prepare their data <i>Data collection:</i> (Meetings) 1) SRC's archeological researchers (all disciplines) 2) By a subgroup of each discipline - Direct observations, verbal and non-verbal communications	<i>Objective:</i> Solving the problem of missing data and a small volume of data <i>Problems:</i> 1) Data had to be prepared to train IA algorithms 2) The operating mode is too structured and involves many participants <i>Means:</i> - Curating the data - Representing the logic of a specific discipline - Building scenarios and cognitive maps <i>Sources:</i> - Archeological researchers: for curating data and expert evaluation for missing data - External repositories for missing features (example: WWF) <i>Data collection:</i> Expert evaluation and external data or repositories for feature engineering	<i>Objective:</i> Testing precise questions crossing disciplines <i>Problems:</i> 1) Intricate questions had to be decomposed 2) Collaboration between archeologists and data scientists became very complex 3) Researchers requested model explainability <i>Means:</i> - Cognitive maps - ML algorithms - Using feature engineering <i>Sources:</i> - Archeological researchers: to link cognitive maps datasets - AI computer specialist: for choosing the best ML algorithms for training and testing <i>Data collection:</i> Data sets validated collectively were used to train and test the designed algorithm	<i>Objective:</i> Linking AI with research to augment organizational knowledge <i>Problems:</i> 1) The immersion part was not yet stabilized, and the link between the AI and the immersion was in-progress 2) COVID period delayed the process 3) The SRC computer scientist died suddenly <i>Means:</i> Deploying the designed algorithm to new datasets (Korean) <i>Sources:</i> 1) Academic literature, 2) Direct day-to-day observations, In-depth interviews, Informal meetings, and discussions <i>Data collection:</i> - Partners' data
Action planning	1) Analyzing the available data and running simple tests with the available algorithms 2) Drawing a cognitive map for each researcher/discipline	1) Meetings were planned for curating data and practicing feature engineering 2) Meetings were planned once a week for six months to build the scenarios - Four to five meetings were planned for each discipline to build the cognitive maps	1) Meetings were planned between archeological researchers individually and AI computer specialists 2) IS specialists act as observers	

Table 1 (continued)

	First cycle (01/17 – 03/17) "The dream of an AI"	Second cycle (04/17 – 12/17) "The AI deception"	Third cycle (01/18 – 04/19) "An IAI to respond to simple questions."	Fourth cycle (05/19 – 11/20) "An integrated AI for researchers"
Action taking	Immediate actions: 1) Canceling the idea of fully integrating data in a centralized dataset 2) Working on knowledge modeling methods (using Vu mapping software) 3) Preparing a guide for data preparation - Analyzing the transfer of individual knowledge, considering tacit and explicit knowledge - Externalization, socialization, internalization, and combination were selected as tools of the focal theory	Immediate actions: - Curating the datasets - For shared datasets, meetings are permitted to curate data - Experts' evaluations were collected for each discipline or sub-discipline, and external data were selected for feature engineering	Immediate actions: - Installation of mixed teams combining one archaeologist and one data scientist - Applying Human-Centric and human-in-the-loop - Approach as instrumental theory - Introduction of post-hoc explainable models such as LIME and SHAP	Immediate actions: - The Project continued remotely - Combining AI learning loop with virtual reality across different disciplines - Extending the scope to a new context (same disciplines but on Korean Data)

researcher said, "When we use the AI, it will tell us exactly how many hunters were needed at that time and how long it took them to kill the 67 reindeer that we discovered in the soil L". The first actions were to analyze the available data type and run simple tests with the available algorithms. For example, some Machine Learning algorithms were used to test and find the humidity indices on a specific level of the historical cave based on the sediment data. However, the results were disappointing because the data was not cleaned and tagged. The team also noticed many missing data with high heterogeneity of the datasets and an absence of a global model that can guide designers in understanding such complex predictions. Therefore, the design team canceled the idea of fully integrating data in a centralized dataset. Instead, two new actions were introduced: 1) work on knowledge modeling methods (using Vu mapping software) and 2) prepare a guide for data preparation. Therefore, the team invited every SRC researcher to use mind-mapping techniques to map his knowledge. Using Vu software and two database design experts helped SRC researchers extract and structure their tacit and explicit knowledge. Cognitive maps were elaborated from "raw data" restricted to one discipline. At the end of this cycle, cognitive maps were designed for each discipline.

4.1.2 Cycle 2: The AI Deception

During the second cycle of the project, the SRC researchers lived a big deception. It started in April 2017 and ended in December 2017. This second cycle aimed to find a solution for the problem of insufficient volume of training data. Indeed, the missing data led to errors in the model's predictions, which generated a general distrust toward AI from the Archaeologists' side. Nevertheless, despite this general disappointment, and during nine months, the SRC researchers worked to build and map two scenarios concerning human prehistorical behavior: one is related to the duration of the sol use, while the other is related to mobility. All the researchers participated in designing the scenarios, as all archaeological disciplines were implicated in the workshops. Scenarios were elaborated from constructs. Cognitive maps contribute to the elaboration of scenarios. During this cycle, IS researchers and data scientists underlined the necessity to prepare data to train IA. The work between archaeologists and data scientists did not work efficiently because it was too structured with many operating methods and involved many participants. Despite these difficulties, the second cycle curated the needed data and introduced new feature engineering methods. Two problems have been resolved: 1) the problem of "missing data" in most data sets and 2) the problem of a small volume of data. These new methods of feature engineering are detailed in the first and the second results of the IAI implementation.

4.1.3 Cycle 3: Using IAI Design to Respond to Precise, Simple Questions

The third cycle started in January 2018 and ended in April 2019. In this cycle, designers trained ML algorithms on simple, precise questions crossing disciplines. Cognitive maps were used to design the tests and choose the ML algorithms. They continued using the two methods (exogenous and endogenous) introduced in the second cycle to solve the missing data problem, and these new methods became well-accepted by archaeologists. Nevertheless, the collaboration between archaeologists and data scientists became very complex and time-consuming because of the high number of participants. Hence, a new work organization was installed based on mixed teams combining one archaeologist and one data scientist. The team replaced structured guides with coordination through conversations. Information Systems (IS) specialists were acting as observers. This cycle also saw the arrival of post-hoc explainable models such as LIME and SHAP. The immersion part was not yet stabilized, and the link between the AI and the immersion was not done. Two new research centers (Spain and Korea) joined the project to test the IAI designed on their data.

4.1.4 Cycle 4: An Integrated AI for Researchers

This fourth and last cycle started in May 2019 and ended in November 2020. This cycle was dedicated to linking AI with research. The idea was to discuss the results of the different IAI tests scientifically and integrate them into the immersion. In some examples, IAI made new predictions and explained to scientists' new results. The arrival of the two new data (from Spain and Korea) improved the external validity of some of the predictions. During the COVID period (March 16th, 2020), the project continued remotely and obtained an extension of 6 months, followed by five additional months. In July 2020, one SRC computer scientist died suddenly, questioning the flow of knowledge between developers (AI and Immersion) and archaeologists.

The transfer of methods happened only with the Korean team, where the IAI design was tested on their data. The Spanish team was not convinced by the method and did not have enough data competencies. The AI / Immersion link was set up with great difficulties and was not fully operational until the end of the project (September 2020). Real exchanges and vibrant discussions related to the results of the IAI tests.

4.2 Research Results: Five Modes of Knowledge Augmentation

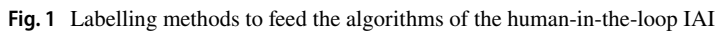
The research results can be divided into two outcomes: the primary and the subsequent outcomes.

The primary outcome of the four-year canonical action research illustrates the importance of the traditional dynamic process of knowledge creation in an organization. SRC's researchers combined their tacit and explicit knowledge through Nonaka's four modes of knowledge conversion: socialization, externalization, combination, and internalization. Indeed, the research findings confirm that SRC researchers from different disciplines organized weekly meetings to exchange their experiences and knowledge related to AI implementation. Each domain benefited from other domains' experiences, remarks, and tacit knowledge. This *socialization* increased the exchange of tacit knowledge between SRC researchers. They were also invited to *externalize the cognitive component* of their tacit knowledge using mind mapping to create new explicit knowledge. Different elements of these maps were *combined* to develop probable scenarios for human prehistorical behavior allowing the operationalization of complex concepts and higher-level concepts such as mobility, duration of occupation, or site function. SRC partners organized many workshops to present explicitly their methods of preparing and curating data. SRC researchers were invited to *internalize* these methods. They were required to prepare their datasets while handling missing data and outliers. They were also trained to identify and remove as many irrelevant and redundant features as possible.

Beyond Nonaka's SECI model, the subsequent outcome of the four-year canonical action research shows that the brute-force methods were ineffective because of the small volume of data available and the deficiency of incomplete or missing data. Results illustrate five new modes of Knowledge Augmentation in organizations that result from the interaction between SRC researchers and AI.

The first new mode of Knowledge Augmentation is **Alimentation**. It is related to how the experts' intuition (opinion) or the *technical component* of SRC researchers' tacit knowledge feeds the training of the Machine Learning (ML) algorithms. Labeling technics were used to aliment the ML algorithms. As shown in Fig. 1, experts' intuitions were uploaded into the datasets to feed the training of the algorithms of the human-in-the-loop IAI.

The second new mode of Knowledge Augmentation is **Aggregation**. The human-in-the-loop IAI was trained using *explicit object-based* knowledge. Indeed, to complete missing data, External References (ER) or external repositories were introduced based on the actualism principle. The actualism principle is *rule-based knowledge* that assumes that the same natural laws and processes in our present-day scientific observations have continuously operated in the past and can be applied anytime. The Aggregation of External References plays an essential role in completing missing attributes. External references were the World Wildlife Fund (WWF) for Fauna and the Köppen for Flora. Indeed,



The human-in-the-loop IAI estimated the labels of unseen observations and presented the properties of the model to

The fourth new mode of Knowledge Augmentation is Reflection. The weekly reflexivity sessions organized by SRC had a double impact on Knowledge Augmentation: from one side, experts helped train the human-in-the-loop IAI; but from another side, the designed AI offered interesting new outcomes that encouraged SNC researchers to start a reflection about the limitations of their methods and their knowledge. As shown in Fig. 5, based on human-in-the-loop IAI predictions, two independent immersive zones were reconstituted: 1) the cave: high-resolution 3D model with a projection of different objects by color according to their nature; 2) the Valley: using IGN fine mesh topography and dynamic procedural landscape plating (Unity 3D). The human-in-the-loop IAI reproduced the Geology, the Flora, and the Fauna of the

External References (Uniformitarian Principle)

A global Database (open source) that aggregates all known species of plants and animals

Prediction of biomes suggested by pollens and fauna in the 52 archaeostratigraphic units of the Arago Caune

Identification of climate trends and paleoenvironmental reconstructions in each level of occupation

BIOME	ECOREGION NAME	ECOREGION CLASS	ORDER DESC	FAMILY	GENUS	SPECIES	Köppen Type	Köppen Region
Palaearctic	Temperate Coniferous Forests	Altai montane forest and forest steppe	140400 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	E (perpet)	T
Palaearctic	Tundra	Bering tundra	410200 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Tundra	Chersky-Kolyma mountain tundra	560700 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Tundra	Chukotka Permafrost tundra	290400 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Coniferous Forests	Da Hinggan-Dzhigaly Mountains conifer	240300 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	W
Palaearctic	Boreal Forests/Taiga	East Siberian taiga	380700 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Deserts and Xeric Shrublands	Great Lakes Basin desert steppe	157200 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	E (perpet)	S
Palaearctic	Tundra	Kamchatka Mountain tundra and forest	113000 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Boreal Forests/Taiga	Kamchatka-Kuril meadows and sparse	140400 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Boreal Forests/Taiga	Kamchatka-Kuril taiga	122000 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Grasslands, Savannas, and Shrublands	Kazakh forest steppe	420300 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Grasslands, Savannas, and Shrublands	Kazakh steppe	860300 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Coniferous Forests	Khangai Mountains conifer forests	2800 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	W
Palaearctic	Tundra	Northwest Siberian coastal tundra	223600 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Boreal Forests/Taiga	Northwest Siberian taiga	112500 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	E (perpet)	T
Palaearctic	Tundra	Northwest Russian-Novaya Zemlya tur	280200 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Boreal Forests/Taiga	Onondaga-Sakhalin taiga	480700 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	W
Palaearctic	Boreal Forests/Taiga	Sakhalin Island taiga	480700 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Montane Grasslands and Shrublands	Sayan-Alpine meadows and tundra	380600 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Grasslands, Savannas, and Shrublands	Sayan-Intermontane steppe	330000 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Coniferous Forests	Sayan montane conifer forests	357800 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Boreal Forests/Taiga	Scandinavian and Russian taiga	213800 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Grasslands, Savannas, and Shrublands	Siberian-Okhon forest steppe	227800 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	W
Palaearctic	Temperate Grasslands, Savannas, and Shrublands	South Siberian forest steppe	340700 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Tundra	Taimyr Central Siberian tundra	560700 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Tundra	Tsagaik-Bald Mountain tundra	217800 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	W
Palaearctic	Boreal Forests/Taiga	Tsagaik-Bald conifer forests	200300 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	W
Palaearctic	Boreal Forests/Taiga	Ural montane forests and tundra	174800 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Broadleaf and Mixed Forests	Ussuri broadleaf and mixed forests	193800 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	W
Palaearctic	Boreal Forests/Taiga	West Siberian taiga	367000 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Temperate Broadleaf and Mixed Forests	Western Siberian hemiboreal forests	220400 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F
Palaearctic	Tundra	Yamal-dudinka tundra	422000 Mammalia Artiodactyla	Cervidae	Rangifer	tarandus	D (continental)	F

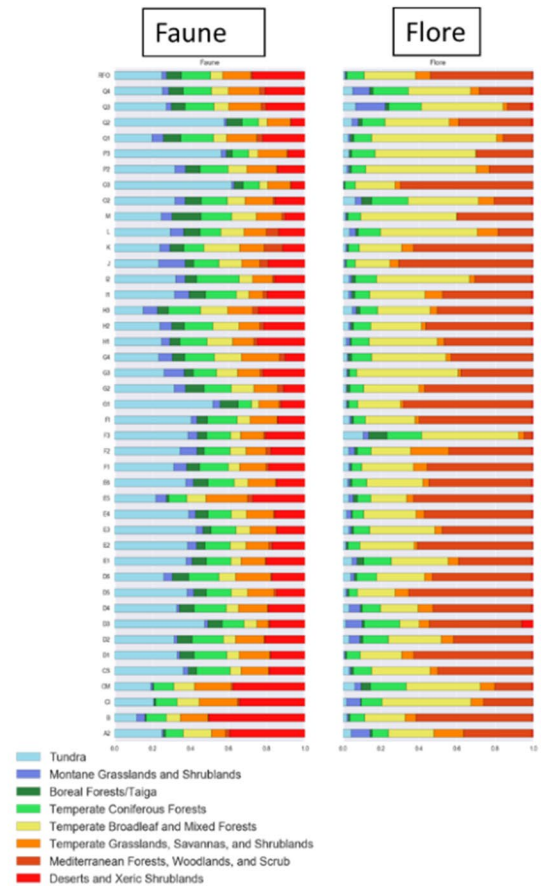


Fig. 2 Aggregation of two External References (the Fauna's WWF and the Flora's Köppen) to train the algorithm

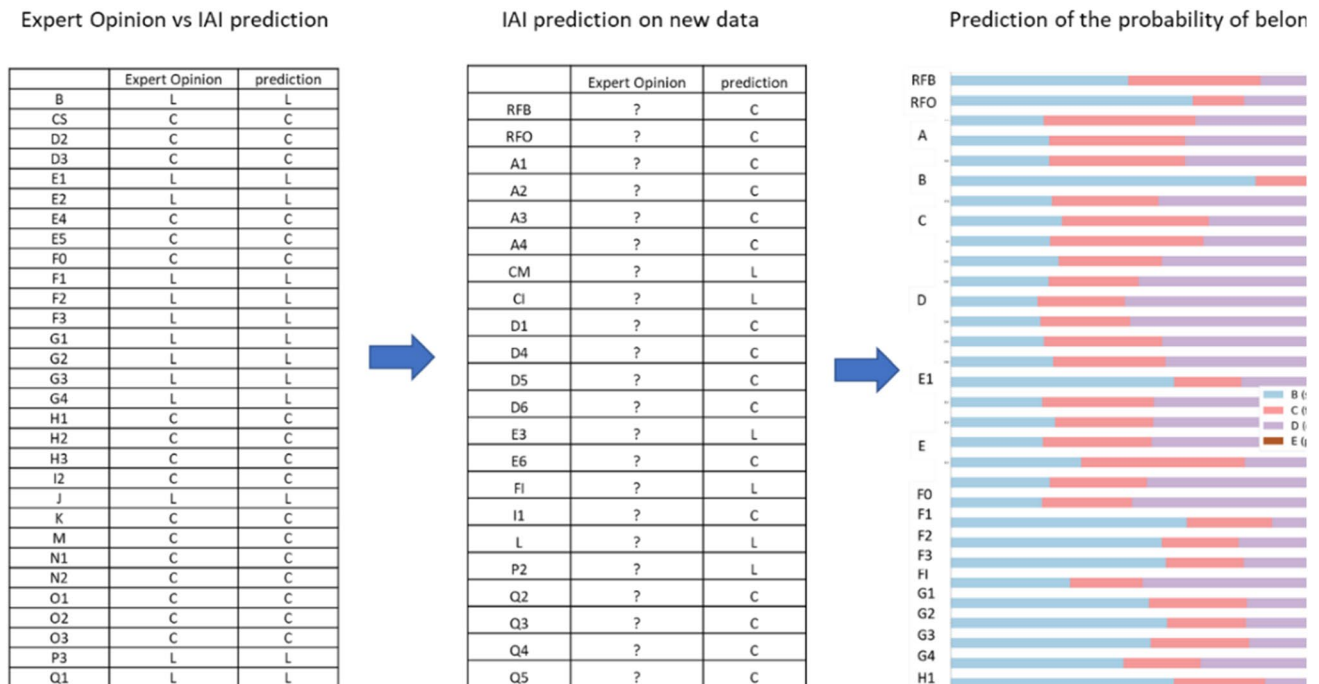
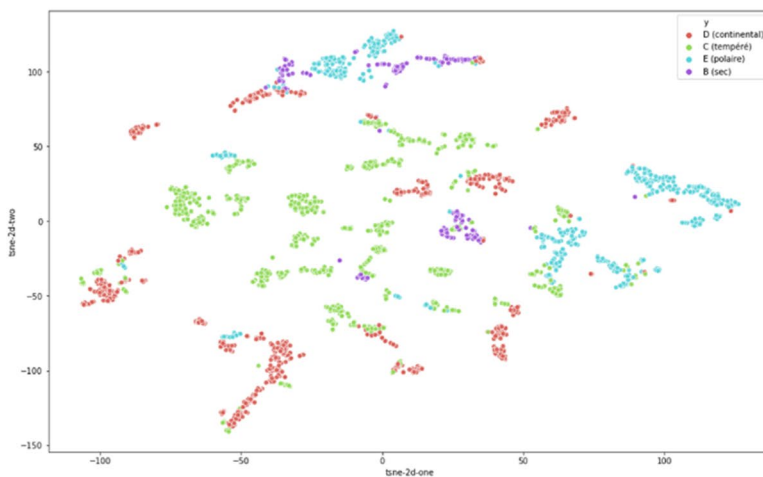


Fig. 3 Informed Artificial Intelligent predictions

Example of temperature across the t-sne components



Example of Shap explanation

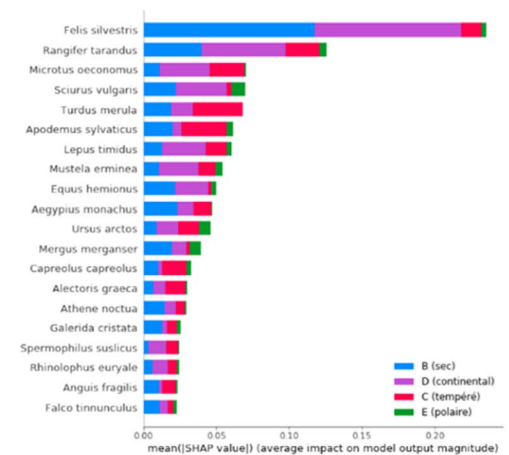


Fig. 4 SHAP and t-SNE used to amplify experts' knowledge through IAI

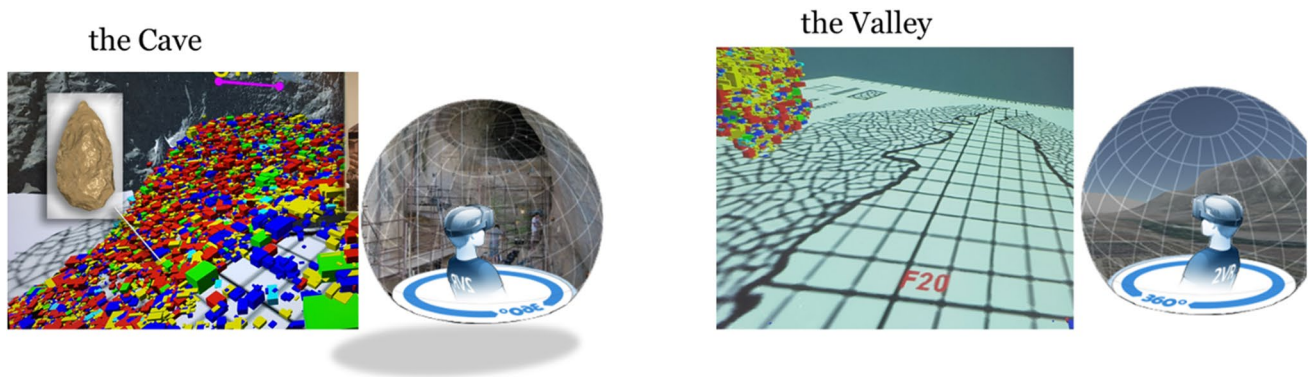


Fig. 5 Researchers experiencing IAI predictive results

Valley. Thanks to virtual reality, SRC researchers experienced IAI's prediction. Reflexivity sessions established a reflexive relation between the human-in-the-loop IAI outputs and researchers' tacit knowledge.

The fifth new mode of Knowledge Augmentation is Fusion. It results from the recursive loops needed to design, train, test, and implement a human-in-the-loop IAI in an organization. The outputs of the first loop of Alimentation, Aggregation, Amplification, and Reflection can set off or trigger a second loop. Once a model is created based on general hypotheses, these learnings can feed more complex models that combine their results to offer more accurate predictions. As shown in Fig. 6, a network of multiple models is connected to solve more complex questions.

5 Discussion, Implications, and Limitations

AI is transforming the way people make decisions. Information Systems scholars have investigated how humans can collaborate with AI to make better decisions (Shrestha et al., 2019). Although research on decision-making emphasizes the importance of knowledge, less work investigates the role of AI in the augmentation of knowledge. This paper contributes to addressing this gap by examining the role of human-in-loop informed AI in augmenting organizational knowledge.

This paper covers a four-year research Project. The strategic starting point of this initiative was that the SRC sought to gain predictability by designing and

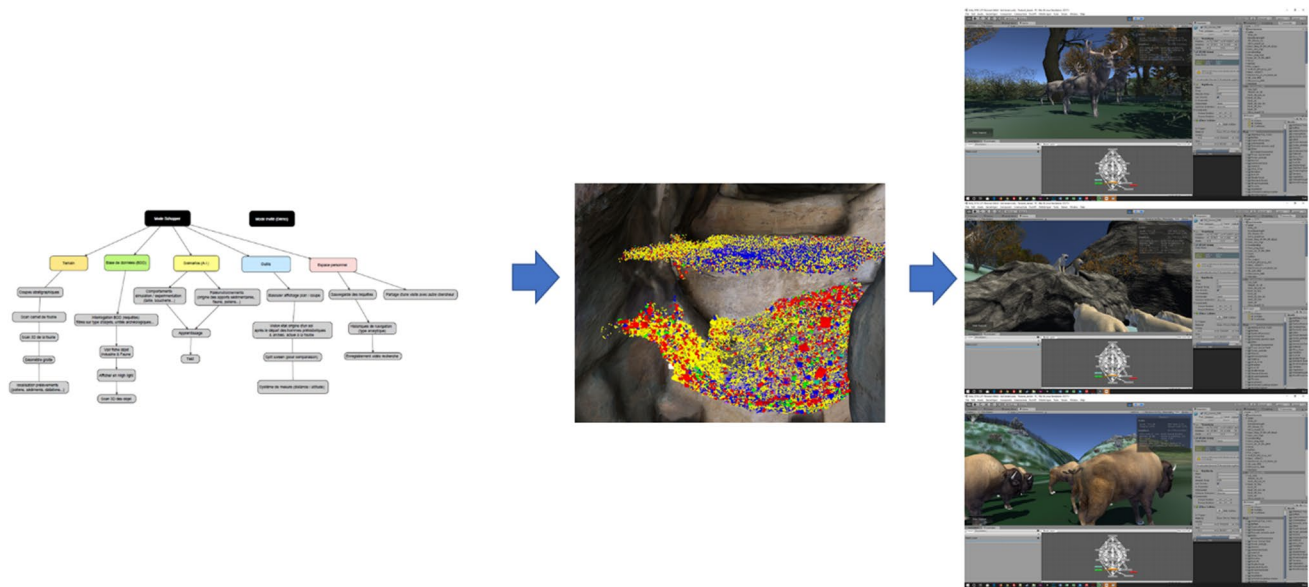


Fig. 6 Multiple algorithms designed to exchange their knowledge and predictions

implementing an AI. The expectation was that the AI would solve the explanation bottleneck that the SRC has lived in since the early 60 s to achieve speed hypothesis testing and accurate predictions. Nevertheless, when the first algorithms were tested, the SRC discovered the need to feed the algorithm with the domain's explicit knowledge and experts' tacit knowledge. A Human-Centric approach was adopted. New tools, techniques, methods, and algorithms were introduced to design a human-in-the-loop Informed AI to address difficulties related to the small volume of data available and the deficiency of incomplete and missing data.

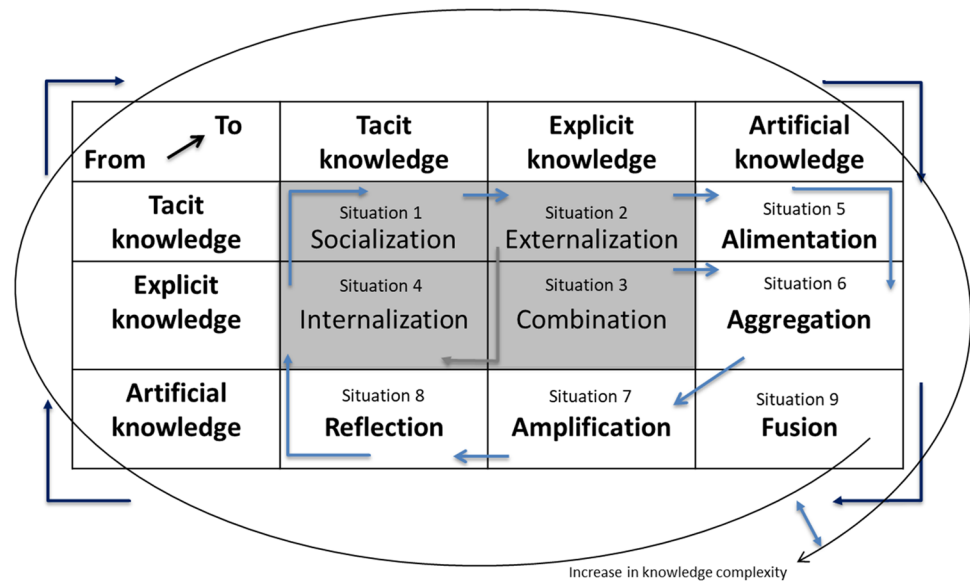
Making decisions by reasoning with knowledge is central to AI and is a key to successful AI implementation (Davis & Marcus, 2015; Russell & Norvig, 2010). Therefore, implicating domain knowledge is essential when designing and implementing an AI (Johnson et al., 2022). This paper provides an example of how organizations can effectively integrate the *cognitive* and *technical* components of their tacit knowledge and object-based and rule-based explicit knowledge while designing and implementing a human-in-the-loop Informed AI. Thus, both Propositions 1 and 2 are confirmed. This result is consistent with previous IS research on the human-in-loop approach, such as the seminal work of Grønsund and Aanestad (2020). The paper provides examples of how designing and implementing an IAI can augment organizational knowledge, which confirms Proposition 3.

As with all research, there are limitations that present interesting directions for future research. First, the researchers' lack of neutrality and the "consulting" style is inherent in action research (Saba & Meissonier, 2020; Baskerville & Wood-Harper, 1996). Future research could adopt

quantitative methods such as a survey. A single case study limits generalisability; however, the rich data gathered from participant observations and informal conversations throughout the four cycles provide deep insights that can be interpreted and applied in other contexts. Another potential limitation of our work is that the observed processes continue to evolve beyond our investigation after the end of the project (Saba & Meissonier, 2020). Future research in other contexts could be conducted to challenge the learning outcomes of this study.

5.1 Implications for Research

Our results contribute to recent IS and organizational literature concerned with the impact of AI implementation on management. The human-in-the-loop Informed AI design illustrates how organizations could answer researchers' call for Human-Centered AI designs (Horvatić & Lipić, 2021; Ramchurn et al., 2021). Our research shows that Nonaka's SECI model is still essential but not anymore sufficient to explain how organizational knowledge can be created in the AI era. Indeed, humans frequently collaborate with AI in significant decision-making situations (Shrestha et al., 2019). Nevertheless, as it is, the SECI model does not consider the AI as an actor in the knowledge creation process. The Human-Centric AI perspective supposes that AI must be designed with the awareness that these systems are part of a more extensive decision-making system consisting of interaction with humans. Addressing the research question, the discussion focuses on how organizational knowledge can be augmented during human and AI interactions. Therefore, the human-in-the-loop Informed AI is added as

Fig. 7 The Knowledge Augmentation Matrix (KAM)

a new element to the SECI model. The Knowledge Augmentation Matrix (KAM) complements and updates the SECI model by crossing three factors: tacit human knowledge, explicit human knowledge, and human-in-the-loop Informed AI. The KAM proposes nine situations: situations 1 to 4 are a reproduction of the SECI model (socialization, externalization, combination, and internalization), and situations 5 to 9 introduce five new modes of Knowledge Augmentation in organizations: 1) Alimentation, 2) Aggregation, 3) Amplification, 4) Reflection, and 5) Fusion. As represented in Fig. 7, the KAM shows that from one side, experts must help in feeding AI with their intuition and knowledge to analyze the data better, acquire the ability to find patterns, select the best responses, and explain their outputs. Alimenting the IAI with experts' tacit intuition is represented by Alimentation, while feeding it with explicit knowledge is called Aggregation. *Alimentation* (situation 5) is a mode of knowledge creation that converges tacit human knowledge into artificial knowledge. *Aggregation* (situation 6) converges explicit knowledge into artificial knowledge.

On another side, interacting with IAI enriches the explicit and tacit knowledge of decision-makers. Two new modes of Knowledge Augmentation can be identified: Amplification and Reflection. *Amplification* (situation 7) converges knowledge created through IAI into explicit human knowledge. Decision-makers can learn from the structured decisions produced by the IAI. This amplification of knowledge is central to human-AI interaction. *Reflection* (situation 8) converges knowledge created through IAI into tacit human knowledge. AI produces interesting new outcomes that can initiate, at the users' level, a reflection on their methods and question their knowledge while experiencing new situations.

Finally, *Fusion* (situation 9) is the mode of Knowledge Augmentation that is produced through 1) the recursive transformation of tacit and explicit knowledge that occurred while feeding the IAI and 2) the enrichment of knowledge that results from the human learning process that occurs during the interaction of human with an IAI. The SECI model explains the transition from individual knowledge to group and organizational knowledge through a complex process. By taking AI into account as a fully-fledged actor in the knowledge creation process, KAM disambiguates the transformation of knowledge from one level to another. Human enriches the artifact, and the artifact infuses human with new knowledge. Through many iterative loops, the process moves from an individual to a collective level, as human-in-the-loop IAI inherently carries a philosophy of sharing and dissemination. The recursive nature of the KAM should be emphasized here, as there is no definite end to the process. The process can be extended indefinitely to integrate more and more knowledge for whatever degree of complexity is required. It is important to stress that the recursive process is not a simple repetition of the previous five components of KAM and that each stage is not merely an addition to the previous stages but instead is driven by the previous one. It is iterative in that the process is repeated with input from the previous application of the process. However, it does not just keep going round and round the loop without any added structure. On the contrary, this recursion leads to a higher level of abstraction that can augment organizational knowledge. Through the KAM, human-in-the-loop IAI does not take the human out of the loop and replace them with algorithms but, on the contrary, augments human involvement in a clear reply to Bailey and Barley (2019) and Markus (2017) calls to keep human in the loop.

5.2 Implications for Practice

Action research is used to construct meaning with practitioners employing an active and interactive method of investigation concerning a practical organizational problem. It installs a problem-solving approach by applying a practical intervention plan. The design of an IAI artifact has contributed to solving an organizational issue related to integrating multidisciplinary knowledge. The project has brought together four partners and six to develop an innovative technological solution centered on studying prehistoric behavior: a Scientific Archaeology Research Center (SRC) with six different disciplines (paleontologist, geologist, etc.), an information Systems laboratory, a start-up specialized in AI and another start-up specialized in virtual reality. The results obtained by the IAI tests were discussed during project meetings, constantly gathering all different disciplines and different partners. Many difficulties in sharing information were observed at all levels: between archaeological disciplines, archaeologists, and data scientists. Experts (archaeologists) from different disciplines were invited to build cognitive maps and develop hypothesis scenarios during collective workshops. These maps and scenarios were then explained to the data scientists, allowing the creation of a shared vision of the research domain and using a common language.

Every single mode of KAM has practical implications. The implication for the practice of Alimentation is presented with the labeling method that encodes Experts' Opinions to feed the algorithm. The Aggregation offers the possibility to train IAI algorithms based on External References to complete missing attributes. When explainable IAI algorithms are implemented, they illustrate the values of the most impactful features and amplify explicit knowledge that a human can quickly learn. Reflection can be implemented in practice by inviting decision-makers to reflexivity sessions dedicated to questioning the limitation of their knowledge and acquiring new experiences and knowledge while interacting with IAI. Finally, the Fusion of new combined outputs in a recursive way can augment the organization's knowledge.

6 Conclusion

Research on AI has generated important insights about the risk of black-box data-driven models and their emerging threats. However, it has not investigated the potential value AI can offer in integrating and enriching multidisciplinary knowledge and augmenting organizational knowledge. Our four-year CAR aiming to introduce an Informed AI is, therefore, a timely contribution to the building and improving understanding of how AI, through Alimentation, increases organizations' ability to train algorithms with tacit human

knowledge. It also shows how explicit knowledge can be aggregated to feed the IAI. While executing some explainable predictions, IAI can amplify organizational knowledge by producing new knowledge graphs and models to help decision-makers learn and acquire new explicit knowledge. The interaction between humans and IAI can intensify researchers' reflexivity by showing the limitations of what they know while experiencing new situations created by the IAI. Through the recursive process, organizational knowledge can continuously be augmented loop after loop. Recursion is inherent to the human learning process.

Concluding, it is essential to underline that this paper is the angular stone toward developing the new theory in IS: the recursive theory of knowledge augmentation in organizations (KAM). Thus, it might be considered a first step toward explaining how AI can enrich multidisciplinary knowledge and develop artificial organizational knowledge. Many future research directions can be proposed based on the paper's results. Researchers can, for example, complement the framework by exploring the different stages that compose the transition from the individual to the collective knowledge creation in interaction with an IAI. Other research can also compare the KAM framework to other AI approaches (Data-driven AI, ...). It is also interesting to compare the results of the KAM framework coming from different fields, such as health, engineering, and education.

Declarations

Conflicts of Interest The authors declare that there are no conflicts of interest.

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